

EXTRA PRIME ATOMS IN POSITIVE SOLUTIONS OF THE EXPLICIT FORMULA: THETA-GROUP CLASS SUMS AND AN OBSTRUCTION

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ABSTRACT. A companion paper studies positive solutions of Weil’s explicit formula with the archimedean data of the Riemann zeta function, and proves that a solution whose zero measure lives on the zeros of ζ plus finitely many extra points, and whose prime measure lives on the interpolation nodes $\log m/4\pi$ of Bondarenko, Radchenko and Seip, is ζ ’s own. By a theorem of Astra in the companion paper, a solution whose zero measure lives on the zeros of ζ , possibly with an atom at the origin, is ζ ’s own whatever its prime measure, so extra prime mass matters only together with extra zeros. This note treats extra atoms of the prime measure off the nodes. Through theta-group class sums we show that an extra atom at a position x with x^2 rational, if it is the only one of this type, is excluded whenever an explicit Gauss-sum invariant $\kappa_{3,2}(x)$ is positive, also in the presence of finitely many extra zeros, and that positions with x^2 irrational are invisible to every argument at leading order in the class sums. Our statements for general non-residue classes and the nowhere-density of the possible positions assume no extra zeros; they remain statements about theta class sums. On the zero side, the same class sums give a positive-spanning property of the basis functions attached to the class 2 mod 3.

Status. This note accompanies the companion paper [4] and has not been peer-reviewed. It records partial results: every numbered statement in Sections 3, 4 and A is proved. The question it addresses, uniqueness when the prime measure has extra atoms, is settled in [4] when there are no extra zeros ([4, Theorem 3.6], due to Astra); the results here matter only when there are extra zeros, and that case remains open. Countably many extra zeros, with summable weights, are also handled in [4] (Section 5). The two numerical observations (Section 6) are not certified and are not used in any proof. The results of [4] that are used are recalled in Section 2, as Facts.

CONTENTS

1. Introduction	2
2. Setting	2
3. Class sums of an extra atom	5
4. Positive spanning by the BRS functions	8
5. Countably many extra zeros: an outline	9
6. Numerical observations	9
7. Open problems	9
AI-use statement	10
Appendix A. Proofs	10
References	13

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1. INTRODUCTION

An *admissible pair* is a pair of positive measures (μ, ν) , μ on the zero side and ν on the prime side beyond $\xi_2 = \log 2/2\pi$, that satisfies Weil's explicit formula with the archimedean data of ζ on a natural test class (Section 2). The companion paper [4] conjectures that every admissible pair, if one exists, is ζ 's own pair p_ζ , and proves this for pairs whose zero measure is carried by the real zeros of ζ together with finitely many extra points and whose prime measure is carried by the set $\mathcal{M} = \{\log m/4\pi : m \geq 1\}$ of the interpolation theorem of Bondarenko, Radchenko and Seip [1] (Fact 2.3). Its proof tests the pair against the interpolation basis of [1], sums the resulting node identities over the class $N \equiv 2 \pmod{3}$, which contains no squares, and concludes with a mean-value argument for the resulting almost periodic function (Fact 2.7).

This note asks what happens when the prime measure has extra atoms off $\mathcal{M}$. An extra atom at η , with $x_\eta = e^{2\pi\eta}$, enters the node identities through the coefficients $B_N(x_\eta)$ of a second modular integral G_{x_η} , and the natural tools are its class sums over a quadratic non-residue class $b \pmod{q}$. We show:

- (i) the class sums have a leading term $\kappa_{q,b}(x)v^{-1/2}$ as $v \rightarrow 0^+$, with $\kappa_{q,b}(x) = 0$ if $x^2 \notin \mathbb{Q}$ and $\kappa_{q,b}(x)$ an explicit combination of normalised Gauss sums if $x^2 \in \mathbb{Q}$ (Proposition 3.2);
- (ii) rational-type extra atoms are constrained: if $\kappa_{q,b}(x_\eta) > 0$ for some admissible class, a lone rational-type extra atom at η is excluded, whatever irrational-type extras accompany it (Theorem 3.4);
- (iii) the positions where $\kappa_{3,2} > 0$ are dense (Lemma 3.5), so the positions that the class sums cannot exclude form a closed, nowhere-dense set (Proposition 3.8);
- (iv) irrational-type extras are invisible at leading order (Proposition 3.6): no argument at leading order in the class sums can exclude them.

Item (iv) is the obstruction. On the zero side, the same class sums show that the values of the BRS functions U_m , $m \equiv 2 \pmod{3}$, at finitely many positive points off Z_ζ positively span (Proposition 4.1); this gives no information on extra prime atoms. Numerical observations suggest that irrational-type extras are visible at intermediate scales (Section 6). The open problems are in Section 7.

The companion paper also contains two theorems contributed by Astra (OpenAI). The first, the zero-side support theorem (Fact 2.4), shows that an admissible pair whose zero measure is carried by $Z_\zeta \cup \{0\}$ is ζ 's pair, with no assumption on the prime measure. So every extra prime mass, atomic or not, at rational or irrational positions, is excluded when there are no extra zeros, and the results of this note matter only when there are extra zeros: there Theorem 3.4(b) and Proposition 3.6 still apply, while Theorem 3.4(a) and Proposition 3.8(a), which assume no extra zeros, are now statements about the class sums only. Lemma 3.3 anticipated the route of that theorem, whose proof carries it out in the limit, for all positions at once. The second theorem extends Fact 2.3 to countably many extra zeros with summable weights and settles the extension proposed in Section 5.

2. SETTING

We recall the definitions and results of [4] that are used; statements quoted from [4] are called Facts.

2.1. Admissible pairs. We write $\widehat{F}(\xi) = \int_{\mathbb{R}} F(t)e^{-2\pi i \xi t} dt$ and $\xi_n = \log n/(2\pi)$ for real $n > 0$. The archimedean functional is $\mathcal{A}(F) := F(\frac{i}{2}) + F(-\frac{i}{2}) + \int_{\mathbb{R}} F \Omega_\infty$, with $\Omega_\infty(t) = \frac{1}{2\pi}(\operatorname{Re} \psi(\frac{1}{4} + \frac{it}{2}) - \log \pi)$. The test class $\mathcal{T} = \bigcup_{0 < \delta < 1/2} \mathcal{T}_\delta$ consists of the even functions F , real on $\mathbb{R}$ and analytic in a neighbourhood of $\{| \operatorname{Im} z | \leq \frac{1}{2} + \delta\}$ for some δ , with $\sup(1 + |z|)^2 |F(z)| < \infty$ there.

Fact 2.1 (Explicit formula on $\mathcal{T}$ [4]; [8], [1, (1.1)]). *For every $F \in \mathcal{T}$, $\mathcal{A}(F) = \sum_{\rho} F(t_{\rho}) + \frac{1}{\pi} \sum_{n \geq 2} \Lambda(n) n^{-1/2} \widehat{F}(\xi_n)$, where $t_{\rho} := -i(\rho - \frac{1}{2})$ and ρ runs over the non-trivial zeros of ζ with multiplicity. No hypothesis on the zeros is made.*

Z_{ζ} is the set of real ordinates γ with $\zeta(\frac{1}{2} + i\gamma) = 0$, m_{γ} the multiplicity, $\mu_{\zeta} = \sum_{\gamma \in Z_{\zeta}} m_{\gamma} \delta_{\gamma}$, $\nu_{\zeta} = \sum_{n \geq 2} \Lambda(n) n^{-1/2} \delta_{\xi_n}$, and $p_{\zeta} = (\mu_{\zeta}, \nu_{\zeta})$. An *admissible pair* is a pair (μ, ν) of positive Radon measures, μ even on $\mathbb{R}$ and ν on $[\xi_2, \infty)$, such that $\mathcal{A}(F) = \mathcal{R}_{\mu, \nu}(F) := \int F d\mu + \frac{1}{\pi} \int \widehat{F} d\nu$ for every $F \in \mathcal{T}$, with absolutely convergent integrals; $\mathcal{K}$ is the set of admissible pairs. We write $\mathcal{M} := \{\log m/(4\pi) : m \geq 1\}$; since $\xi_n = \log(n^2)/(4\pi)$, the prime-power points $\xi_{\text{PP}} = \{\xi_n : n \in \text{PP}\}$ lie in $\mathcal{M}$.

$\vartheta(\tau) = \sum_{n \in \mathbb{Z}} e^{\pi i n^2 \tau}$ is Jacobi's theta function and Γ_{ϑ} the theta group, generated by $S = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$ and $T^2 = \begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix}$ and viewed, as in [1], in $\text{PSL}_2(\mathbb{R})$, so that matrices are identified up to sign and $S^2 = 1$; we write $\tau = \sigma + i\nu \in \mathbb{H}$. μ is the Möbius function, $\mathbf{1}_{\square}(m) = 1$ if m is a perfect square and 0 otherwise, and $\Lambda(\sqrt{m}) := 0$ if m is not a square.

2.2. The interpolation basis. For $s \in \mathbb{C}$ let $F(\tau, s) = \sum_{N \geq 0} \alpha_N(s) e^{\pi i N \tau}$ be the unique 2-periodic analytic function of moderate growth on $\mathbb{H}$ satisfying

$$(2.1) \quad F(\tau, s) + (\tau/i)^{-1/2} F(-1/\tau, s) = \mathfrak{p}_s(\tau) := (\tau/i)^{-s} + (\tau/i)^{s-1/2}$$

(principal branches); this is $F_{1/2}^-(\tau, s)$ of [1, Theorem 3.1]. Its coefficients grow polynomially in N , are entire in s , and are real on the line $\text{Re } s = \frac{1}{4}$. Let b_N ($N \geq 0$) be the even Fourier eigenfunctions with eigenvalue $+1$ of Radchenko and Viazovska [7] (the functions b_N^- of [1, §3], cf. [1, §3.4.1]), so that $b_N(\sqrt{n}) = \delta_{Nn}$ for $n \geq 1$, and put $B_N(x) := \sum_{k \geq 1} b_N(kx)$ for $x > 0$. By [1, Theorem 1.1], which assumes neither RH nor simplicity of the zeros, there are even entire functions U_m ($m \geq 1$) and $V_{\rho, j}$ (ρ a non-trivial zero with $\text{Im } \rho > 0$, $0 \leq j < m(\rho)$) with $U_m^{(j)}(t_{\rho}) = 0$, $\widehat{U_m}(\log m'/4\pi) = \delta_{mm'}$, $V_{\rho, j}^{(j')}(t_{\rho'}) = \delta_{(\rho, j), (\rho', j')}$ and $\widehat{V_{\rho, j}}(\log m/4\pi) = 0$; their real and imaginary parts lie in $\mathcal{T}$. These are the *BRS functions*. With $\Xi(t) := \xi(\frac{1}{2} + it)$, $\xi(s) = \frac{1}{2}s(s-1)\Gamma_{\mathbb{R}}(s)\zeta(s)$ and $\Gamma_{\mathbb{R}}(s) = \pi^{-s/2}\Gamma(\frac{s}{2})$, explicitly [1, §§4.2–4.3]

$$(2.2) \quad U_m(z) = \frac{m^{-1/4}}{2\pi} h_m(\tfrac{1}{2} + iz), \quad h_m(s) = \frac{\Gamma_{\mathbb{R}}(s)\zeta(s)}{2} \sum_{d^2|m} \mu(d) \alpha_{m/d^2}(s/2).$$

and $\widehat{U_m}(\xi) = m^{-1/4} e^{\pi \xi} u_m(e^{2\pi \xi})$ with $u_m(x) = \sum_{d^2|m} \mu(d) B_{m/d^2}(x)$. By the explicit formula, applied to real and imaginary parts,

$$(2.3) \quad \mathcal{A}(U_m) = \frac{1}{\pi} \Lambda(\sqrt{m}) m^{-1/4}, \quad \mathcal{A}(V_{\rho, 0}) = 2m(\rho), \quad \mathcal{A}(V_{\rho, j}) = 0 \quad (j \geq 1),$$

where $m(\rho)$ is the multiplicity of ρ and $\Lambda(\sqrt{m}) := 0$ if m is not a square [4]. Indeed U_m vanishes at every t_{ρ} and $\widehat{U_m}(\xi_n) = \widehat{U_m}(\log(n^2)/4\pi) = \delta_{m, n^2}$; for $V_{\rho, j}$ the prime sum vanishes, and the zeros with negative imaginary part contribute as their conjugates do, by evenness.

Fix a finite symmetric set $E \subset \mathbb{R} \setminus Z_{\zeta}$, and put $E_+ := E \cap [0, \infty)$ and $s_e := \frac{1}{4} + \frac{ie}{2}$. Let $\mathcal{X} \subset [\xi_2, \infty) \setminus \mathcal{M}$ be finite, and write $x_{\eta} := e^{2\pi \eta}$ for $\eta \in \mathcal{X}$; so $x_{\eta} \geq 2$ and $x_{\eta}^2 \notin \mathbb{N}$.

Lemma 2.2 (Structure of pairs with extra atoms). *Let μ be an even real measure carried by $Z_{\zeta} \cup E$ and ν a real measure carried by $\mathcal{M} \cup \mathcal{X}$ such that $\mathcal{A}(F) = \mathcal{R}_{\mu, \nu}(F)$ for $F = U_m$ ($m \geq 1$) and $F = V_{\rho, j}$. Write $a_e := \mu(\{e\})$, $w_{\gamma} := \mu(\{\gamma\})$, $\beta_{\eta} := \nu(\{\eta\})$, $\ell_m := m^{1/4} \nu(\{\frac{\log m}{4\pi}\})$, $c_N := \sum_{k^2|N} \ell_{N/k^2}$,*

$$\tilde{\varrho}_e := \frac{a_e \Xi(e)}{\frac{1}{4} + e^2} \quad (e > 0), \quad \tilde{\varrho}_0 := 2a_0 \Xi(0), \quad c_N(E) := \sum_{e \in E_+} \tilde{\varrho}_e \alpha_N(s_e) \in \mathbb{R}.$$

Then for every $N \geq 1$

$$(2.4) \quad C_N = c_N(E) + \frac{1}{2}(\log N) \mathbf{1}_{\square}(N) - \sum_{\eta \in \mathcal{X}} \beta_{\eta} \sqrt{x_{\eta}} B_N(x_{\eta}).$$

If $\mathcal{X} = \emptyset$, moreover: $c_1 = c_2 = c_3 = 0$ when ν is carried by $[\xi_2, \infty)$; $w_{\gamma} = m_{\gamma} - \frac{1}{2} \sum_{e \in E} a_e V_{\rho,0}(e)$ for $\gamma > 0$ and $\rho = \frac{1}{2} + i\gamma$; $\frac{1}{2} \sum_e a_e V_{\rho,0}(e) = m(\rho)$ for every zero ρ off the line with $\text{Im } \rho > 0$; $\sum_e a_e V_{\rho,j}(e) = 0$ for $1 \leq j < m(\rho)$; and $(\mu, \nu) = p_{\zeta}$ if and only if $a = 0$, if and only if $c(E) \equiv 0$. If $(\mu, \nu) \in \mathcal{K}$, then $C_N \geq 0$ for all N ; in particular $c_N(E) \geq 0$ for every non-square N when $\mathcal{X} = \emptyset$.

The proof is in Appendix A.1. The case $\mathcal{X} = \emptyset$, with the statements that follow “moreover”, is [4, Lemma 3.7], which allows countable E .

Fact 2.3 (Finitely many extra zeros; the case of finite E of [4, Theorem 3.8]). *Let $E \subset \mathbb{R} \setminus Z_{\zeta}$ be finite and symmetric. Let μ be an even real measure carried by $Z_{\zeta} \cup E$ and ν a real measure carried by $\mathcal{M}$ such that $\mathcal{A}(F) = \int F d\mu + \frac{1}{\pi} \int \hat{F} d\nu$ for every BRS function F (in particular, if the identity holds on $\mathcal{T}$ with absolutely convergent integrals). Assume $\nu(\{\frac{\log m}{4\pi}\}) \geq 0$ for every $m \equiv 2 \pmod{3}$, and $\mu(\{0\}) \geq 0$ if $0 \in E$. Then RH holds, $\mu = \mu_{\zeta}$ and $\nu = \nu_{\zeta}$. In particular every admissible pair carried by $(Z_{\zeta} \cup E) \times \mathcal{M}$ is p_{ζ} ([4, Theorem B(b)]).*

In [4, Theorem 3.8] this is proved, by a theorem of Astra, for countable symmetric $E \subset \mathbb{R} \setminus Z_{\zeta}$ with $\sum_{e \in E, e > 0} |\mu(\{e\})| |\zeta(\frac{1}{2} + ie)| (1 + e)^{-1/2} < \infty$, or more generally with partial sums of $|\mu(\{e\})| |\zeta(\frac{1}{2} + ie)|$ over $0 < e \leq T$ of order $o(T^{1/2})$.

Fact 2.4 (Zero-side support; [4, Theorem 3.6], due to Astra). *If $(\mu, \nu) \in \mathcal{K}$ and μ is carried by $Z_{\zeta} \cup \{0\}$, then RH holds and $(\mu, \nu) = p_{\zeta}$. No assumption on ν is made.*

2.3. The theta function near rational points. A fraction a/c in lowest terms ($c \geq 1$) has one of the three *types* (even, odd), (odd, even) and (odd, odd), according to the parities of a and c . For $a/c \in \mathbb{Q}$ in lowest terms let $g(a/c) := c^{-1} \sum_{n \bmod c} e^{\pi i a n^2 / c}$ if ac is even, and $g(a/c) := 0$ if ac is odd. We use $\vartheta(-1/\tau) = (\tau/i)^{1/2} \vartheta(\tau)$, $\vartheta(\tau + 2) = \vartheta(\tau)$, and the fact that ϑ has no zeros in $\mathbb{H}$ (see [1, §3.2]). Principal branches are used throughout; $(\tau/i)^{\alpha} > 0$ for $\tau \in i(0, \infty)$.

Lemma 2.5 (ϑ near real points). (a) *Let $r = a/c \in \mathbb{Q}$ in lowest terms, $c \geq 1$. For $z \in \mathbb{H}$ with $Y := \text{Im}(-1/z) \geq 1$, $\vartheta(r+z) = g(r)(z/i)^{-1/2} + O_r(|z|^{-1/2} e^{-\pi Y/(4c^2)})$. In particular, if $z \rightarrow 0$ with $|\text{Re } z| \leq M \text{Im } z$, then $\vartheta(r+z) = g(r)(z/i)^{-1/2} + O(|z|^{-1/2} e^{-c'/|z|})$ for some $c' = c'(r, M) > 0$.*

(b) *Let $r \in \mathbb{R} \setminus \mathbb{Q}$. If $\varepsilon \rightarrow 0^+$ and $|\delta| \leq M\varepsilon^2$, then $\vartheta(r + \delta + i\varepsilon) = o(\varepsilon^{-1/2})$.*

(c) *For $r \in \mathbb{Q} \setminus \{0\}$, $g(-1/r) = |r|^{-1/2} e^{-i\pi \text{sgn}(r)/4} g(r)$.*

The proof is in Appendix A.2.

2.4. Cusp expansions. Let $\chi : \Gamma_{\vartheta} \rightarrow \{\pm 1\}$ be the character with $\chi(S) = -1$ and $\chi(T^2) = 1$; it is well defined because S and T^2 generate Γ_{ϑ} with the single relation $S^2 = 1$ [1, §3.1]. For $f : \mathbb{H} \rightarrow \mathbb{C}$ and $\gamma \in \Gamma_{\vartheta}$ put

$$(f|\gamma)(\tau) := \chi(\gamma) \frac{\vartheta(\tau)}{\vartheta(\gamma\tau)} f(\gamma\tau),$$

a right action: $(f|\gamma_1)|\gamma_2 = f|(\gamma_1\gamma_2)$. As $\vartheta(\tau)/\vartheta(-1/\tau) = (\tau/i)^{-1/2}$, the functional equations (2.1) and (3.1) and periodicity read $f - f|S = \mathfrak{p}$ and $f|T^2 = f$, for $(f, \mathfrak{p}) = (F(\cdot, s), \mathfrak{p}_s)$ and

$(G_x, \mathfrak{p}_x)$. If $\gamma = g_m \cdots g_1$ with $g_k \in \{S, T^{\pm 2}\}$ and $w_k := g_k \cdots g_1$ ($w_0 = 1$), then telescoping $f|w_{k-1} - f|w_k = (f - f|g_k)|w_{k-1}$ gives

$$(2.5) \quad f = f|\gamma + \sum_{k: g_k=S} \mathfrak{p}|w_{k-1}.$$

For the cusp $\frac{2}{3}$ we use $\gamma = ST^2ST^2S = \begin{pmatrix} -2 & 1 \\ 3 & -2 \end{pmatrix}$, which maps $\frac{2}{3}$ to ∞ , with S in positions $k = 1, 3, 5$ and $w_2 = T^2S$, $w_4 = T^2ST^2S = \begin{pmatrix} 3 & -2 \\ 2 & -1 \end{pmatrix}$. For $\tau = \frac{2}{3} + iv$ one finds $T^2S\tau = 2 - 1/\tau \rightarrow \frac{1}{2}$, $w_4\tau = 9iv/(1 + 6iv)$ and $\gamma\tau = -\frac{2}{3} + \frac{i}{9v}$; moreover $\chi(\gamma) = -1$, $\chi(T^2S) = -1$, $\chi(w_4) = 1$,

$$(2.6) \quad \frac{\vartheta(\tau)}{\vartheta(T^2S\tau)} = (\tau/i)^{-1/2}, \quad \frac{\vartheta(\tau)}{\vartheta(w_4\tau)} = (\tau/i)^{-1/2}((2 - 1/\tau)/i)^{-1/2},$$

and both are analytic near $\tau = \frac{2}{3}$, the second with value $(\frac{3}{2})^{1/2}e^{i\pi/4} \cdot \sqrt{2}e^{i\pi/4} = i\sqrt{3}$ there.

Fact 2.6 (The class 2 mod 3; [4, Lemma B.2]). *Let $\operatorname{Re} s = \frac{1}{4}$. As $v \rightarrow 0^+$,*

$$\begin{aligned} & \frac{1}{3} \left[F(iv, s) + \omega F(\frac{2}{3} + iv, s) + \bar{\omega} F(\frac{4}{3} + iv, s) \right] \\ &= \frac{1}{3} (1 - 3 \cdot 9^{-s}) v^{-s} + \frac{1}{3} (1 - 3 \cdot 9^{s-1/2}) v^{s-1/2} + c(s) + O_s(v^{3/4}), \end{aligned}$$

with a constant $c(s)$. Consequently, if $E_+ \subset [0, \infty)$ is finite, $\tilde{\varrho}_e \in \mathbb{R}$ and $c_N := \sum_{e \in E_+} \tilde{\varrho}_e \alpha_N(s_e)$, then

$$\Theta_E^{(2)}(v) := \sum_{N \equiv 2(3)} c_N e^{-\pi N v} = \sum_{e \in E_+, e > 0} (B_e v^{-s_e} + \overline{B_e} v^{-\overline{s_e}}) + B_0 v^{-1/4} + C + O(v^{3/4}),$$

with $B_e = \frac{\tilde{\varrho}_e}{3} (1 - 3 \cdot 9^{-s_e})$ for $e > 0$, $B_0 = \frac{2\tilde{\varrho}_0}{3} (1 - \sqrt{3})$ (and $B_0 = 0$ if $0 \notin E_+$), and a real constant C . In particular $\Theta_E^{(2)}(v) = O(v^{-1/4})$.

The function $\Theta_E^{(2)}$ is the class sum of the node identities of Lemma 2.2 with $\mathcal{X} = \emptyset$ over the class 2 mod 3; in [4] its expansion and a mean-value argument give Fact 2.3.

Fact 2.7 (Mean-value step; the case of finite E_+ of [4, Lemma B.4]). *Let $E_+ \subset [0, \infty)$ be finite, let $a_e \in \mathbb{R}$ ($e \in E_+$), put $\tilde{\varrho}_e := a_e \Xi(e)/(\frac{1}{4} + e^2)$ for $e > 0$ and $\tilde{\varrho}_0 := 2a_0 \Xi(0)$, and $c_N := \sum_{e \in E_+} \tilde{\varrho}_e \alpha_N(s_e)$. If $a_0 \geq 0$ when $0 \in E_+$, and $c_N \geq 0$ for every $N \equiv 2 \pmod{3}$, then $\tilde{\varrho}_e = 0$ for every $e \in E_+$.*

3. CLASS SUMS OF AN EXTRA ATOM

We turn to admissible pairs whose prime measure has atoms off $\mathcal{M}$, keeping the notation of Lemma 2.2. An extra atom at η enters the node identities (2.4) through the numbers $B_N(x_\eta)$, and these are the Fourier coefficients of a second modular integral.

Definition 3.1 (Class sums of an extra atom). For $x > 0$ and $\tau \in \mathbb{H}$ put $G_x(\tau) := \sum_{N \geq 0} B_N(x) e^{\pi i N \tau}$. This is the modular integral $F_{1/2}^-(\tau, \phi_x)$ of [1, Theorem 3.1] attached to $\phi_x(\tau) := \frac{1}{2}(\vartheta(x^2\tau) - 1)$ (cf. the proof of [1, Theorem 1.1], where the argument of ϑ is printed as zx ; the square is needed, since $B_N(\sqrt{m}) = \#\{k \geq 1 : k^2 m = N\}$ for integers $m \geq 1$): it is 2-periodic, of moderate growth, and

$$(3.1) \quad G_x(\tau) + (\tau/i)^{-1/2} G_x(-1/\tau) = \mathfrak{p}_x(\tau) := \frac{1}{2} [\vartheta(x^2\tau) - 1 + x^{-1} \vartheta(\tau/x^2) - (\tau/i)^{-1/2}].$$

For an odd prime q , a quadratic non-residue $b \pmod{q}$ and $v > 0$ let

$$T_{q,b}^{(x)}(v) := \sum_{N \equiv b(q)} B_N(x) e^{-\pi N v} = \frac{1}{q} \sum_{j \pmod{q}} e^{-2\pi i j b/q} G_x(\frac{2j}{q} + iv).$$

Recall that the number $g(r)$ of Section 2.3 describes ϑ near the rational point r : $\vartheta(r+z) = g(r)(z/i)^{-1/2} + O(|z|^{-1/2}e^{-c/|z|})$ as $z \rightarrow 0$ non-tangentially in $\mathbb{H}$ (Lemma 2.5).

Proposition 3.2 (Leading term of non-residue class sums). *Let $x > 0$, q an odd prime and b a quadratic non-residue mod q . As $v \rightarrow 0^+$,*

$$T_{q,b}^{(x)}(v) = \kappa_{q,b}(x) v^{-1/2} + o(v^{-1/2}),$$

where $\kappa_{q,b}(x) = 0$ if $x^2 \notin \mathbb{Q}$, and for $x^2 \in \mathbb{Q}$, $\kappa_{q,b}(x)$ is an explicit finite linear combination of values of g , the error being $O(1)$. For $(q,b) = (3,2)$, with $X := x^2 \in \mathbb{Q}$ and $\omega := e^{2\pi i/3}$,

$$\kappa_{3,2}(x) = \frac{2}{3} \operatorname{Re}(\omega K), \quad K = \frac{g(\frac{2X}{3})}{2x} + \frac{e^{i\pi/4}}{\sqrt{6}x} \left[g\left(-\frac{3X}{2}\right) - g\left(\frac{X}{2}\right) \right] - \frac{i g(-2X)}{2\sqrt{3}x}.$$

For $X = m \in \mathbb{N}$ this gives $\kappa_{3,2}(x) = 1/(3x)$ if $m \equiv 2 \pmod{3}$ and $\kappa_{3,2}(x) = 0$ otherwise.

The proof, in Appendix A.3, unrolls (3.1) along a reduction word at each cusp $2j/q$. The contribution $(B_0(x) - \frac{1}{2x})\vartheta(\tau)$ that a single atom would make to every class appears at each cusp with the same coefficient, because the multiplier of the reduction word is -1 at every cusp with odd denominator; it therefore cancels exactly in every non-residue class, since $\sum_j e^{-2\pi i j b/q} \vartheta(\frac{2j}{q} + iv) = q \sum_{n^2 \equiv b(q)} e^{-\pi n^2 v} = 0$. What remains samples ϑ near the points $x^2 \rho$ and ρ/x^2 for finitely many rationals ρ . The case $X = m \in \mathbb{N}$ is a check: then $B_N(\sqrt{m}) = \sum_k \delta_{N,k^2 m}$, so $G_x = \phi_x$ and $T_{3,2}^{(x)}(v) = \sum_{3 \nmid k, k^2 m \equiv 2(3)} e^{-\pi k^2 m v}$.

Lemma 3.3 (Node-positive test functions suffice). *Let $(\mu, \nu) \in \mathcal{K}$ be carried by $(Z_\zeta \cup E) \times (\mathcal{M} \cup \mathcal{X})$ as in Lemma 2.2, and let $\eta_0 \in \mathcal{X}$. Suppose some $F \in \mathcal{T}$ (or the real part of an even complex F whose real and imaginary parts lie in $\mathcal{T}$) vanishes at every t_ρ and on E , has $\hat{F} \geq 0$ on $(\mathcal{M} \cap [\xi_2, \infty)) \cup \mathcal{X}$, $\hat{F} = 0$ on ξ_{PP} , and $\hat{F}(\eta_0) > 0$. Then $\beta_{\eta_0} = 0$.*

Proof. By the explicit formula (Fact 2.1), $\mathcal{A}(F) = 0$. On the spectral side $\int F d\mu = 0$ and $\frac{1}{\pi} \int \hat{F} d\nu \geq \frac{1}{\pi} \beta_{\eta_0} \hat{F}(\eta_0) \geq 0$. $\square$

No positivity of F on $\mathbb{R}$, and none of $\hat{F}$ off $\mathcal{M} \cup \mathcal{X}$, is needed. Examples are $\Xi^2 H$ and non-negative combinations of BRS functions U_m over non-square m . When $E \subseteq \{0\}$ the route of this lemma is carried out, in the limit and for all positions at once, by the proof of Fact 2.4.

Theorem 3.4 (Rational prime-side extras). *Let $(\mu, \nu) \in \mathcal{K}$ with μ carried by $Z_\zeta \cup E$ and ν carried by $\mathcal{M} \cup \mathcal{X}$, where E and $\mathcal{X}$ are finite as in Lemma 2.2. Let $\mathcal{X}_\mathbb{Q} := \{\eta \in \mathcal{X} : x_\eta^2 \in \mathbb{Q}\}$.*

- (a) *If $E = \emptyset$, then for every odd prime q and every quadratic non-residue $b \pmod{q}$,*

$$\sum_{\eta \in \mathcal{X}_\mathbb{Q}} \beta_\eta \sqrt{x_\eta} \kappa_{q,b}(x_\eta) \leq 0.$$
- (b) *For arbitrary finite E , the inequality holds for $(q,b) = (3,2)$.*

Consequently, if some non-negative combination of the $\kappa_{q,b}$ admissible in (a) or (b) is positive at every x_η , $\eta \in \mathcal{X}_\mathbb{Q}$, then $\beta_\eta = 0$ for all $\eta \in \mathcal{X}_\mathbb{Q}$. In particular, if η is the only rational-type extra atom and $\kappa_{q,b}(x_\eta) > 0$ for one admissible (q,b) , then $\beta_\eta = 0$, whatever irrational-type extras accompany it. (With several rational-type extras the inequalities of (a) and (b) are signed sums, and individual terms may have opposite signs.)

Proof. A non-residue class contains no squares, so by (2.4) and Lemma 2.2, $0 \leq C_N = c_N(E) - \sum_\eta \beta_\eta \sqrt{x_\eta} B_N(x_\eta)$ for $N \equiv b \pmod{q}$. Multiply by $e^{-\pi N v}$ and sum:

$$0 \leq \sum_{N \equiv b(q)} c_N(E) e^{-\pi N v} - \sum_\eta \beta_\eta \sqrt{x_\eta} T_{q,b}^{(x_\eta)}(v).$$

The first sum vanishes if $E = \emptyset$; for $(q, b) = (3, 2)$ it is $O(v^{-1/4})$ by Fact 2.6. Multiply by $v^{1/2}$ and let $v \rightarrow 0^+$; by Proposition 3.2, $v^{1/2}T_{q,b}^{(x_\eta)}(v) \rightarrow \kappa_{q,b}(x_\eta)$, which is 0 unless $\eta \in \mathcal{X}_\mathbb{Q}$. For the consequence, if $\lambda_{q,b} \geq 0$ and $\sum \lambda_{q,b} \kappa_{q,b}(x_\eta) > 0$ for $\eta \in \mathcal{X}_\mathbb{Q}$, then $0 \geq \sum_\eta \beta_\eta \sqrt{x_\eta} \sum \lambda_{q,b} \kappa_{q,b}(x_\eta)$ is a sum of non-negative terms, so each vanishes. $\square$

For $E \neq \emptyset$ and $q \geq 5$ the argument needs the analogue of Fact 2.6 for the class $b \bmod q$, which is not proved here (Section 7). For $E = \emptyset$, Fact 2.4 gives more than part (a): every β_η vanishes. Part (a) remains a statement about the class sums.

Lemma 3.5 (A closed form on a dense family). *Let $d \equiv 1 \pmod{24}$ be prime, let $a \equiv 5 \pmod{6}$ be an integer with $d \nmid a$, and put $X := a/d$, $x := \sqrt{X}$. Then*

$$x \kappa_{3,2}(x) = \frac{(a/d)}{3\sqrt{d}},$$

where (a/d) is the Legendre symbol. The set of such X with $(a/d) = +1$ is dense in $(0, \infty)$; hence $\{X \in \mathbb{Q}_{>0} : \kappa_{3,2}(\sqrt{X}) > 0\}$ is dense.

The proof (Appendix A.4) evaluates the four values of g in Proposition 3.2 by quadratic Gauss sums, and obtains the density from the Pólya–Vinogradov inequality.

Proposition 3.6 (Irrational extras are invisible at leading order). *If $x^2 \notin \mathbb{Q}$, then $T_{q,b}^{(x)}(v) = o(v^{-1/2})$ as $v \rightarrow 0^+$, for every odd prime q and every non-residue $b \bmod q$. Consequently an extra atom η with $x_\eta^2 \notin \mathbb{Q}$ makes no contribution to the inequalities of Theorem 3.4.*

Proof. This is the irrational case of Proposition 3.2. $\square$

The reading of this proposition is an obstruction: no argument at leading order in the class sums can exclude irrational-type extras. Shifted prime powers $x^2 = n^2 \pm \delta$, and the spurious points of near-optimal test functions observed numerically, are of this type. Heuristically, such extras are visible at intermediate scales: near a good rational approximation of x^2 the class sum behaves like that of the rational point (Numerical observation 6.2).

Lemma 3.7 (Continuity of G_x in x). *For fixed $\tau \in \mathbb{H}$, the map $x \mapsto G_x(\tau)$ is continuous on $(0, \infty)$.*

The proof is in Appendix A.5. It uses the contour-integral construction of the modular integral in [1, §3.3], dominated convergence on a fundamental domain, and the transport of continuity by the cocycle relation.

Proposition 3.8 (Lone extras sit on a nowhere-dense set). *Let $\mathfrak{B}$ be the set of $X \in (4, \infty) \setminus \mathbb{Z}$ such that $T_{q,b}^{(\sqrt{X})}(v) \leq 0$ for every $v > 0$, every odd prime q and every non-residue $b \bmod q$.*

- (a) *If $(\mu, \nu) \in \mathcal{K}$ with μ carried by Z_ζ and ν carried by $\mathcal{M} \cup \{\eta\}$, where $\eta \notin \mathcal{M}$ and $\nu(\{\eta\}) > 0$, then $x_\eta^2 \in \mathfrak{B}$.*
- (b) *$\mathfrak{B}$ is closed relative to $(4, \infty) \setminus \mathbb{Z}$, and nowhere dense in $\mathbb{R}$.*

By Fact 2.4, the hypothesis of (a) never holds; (b) is a statement about theta class sums.

Proof. (a) Here $x_\eta \geq 2$ and $x_\eta^2 \notin \mathbb{N}$, so $X := x_\eta^2 \in (4, \infty) \setminus \mathbb{Z}$. With $E = \emptyset$ and $\mathcal{X} = \{\eta\}$, (2.4) gives $0 \leq C_N = -\beta_\eta \sqrt{x_\eta} B_N(x_\eta)$ at every non-square N , so $B_N(x_\eta) \leq 0$ for all N in every non-residue class, and $T_{q,b}^{(x_\eta)}(v) \leq 0$. (b) For fixed v, q, b , the map $X \mapsto T_{q,b}^{(\sqrt{X})}(v)$ is a finite combination of values $G_{\sqrt{X}}(\tau)$, hence continuous by Lemma 3.7; so $\mathfrak{B}$ is an intersection of relatively closed sets. If $\kappa_{q,b}(\sqrt{X}) > 0$ for some (q, b) , then $X \notin \mathfrak{B}$ by Proposition 3.2. If the closure of $\mathfrak{B}$ contained an open interval I , then $I \subset [4, \infty)$ and, by relative closedness, $I \setminus \mathbb{Z} \subset \mathfrak{B}$; but by Lemma 3.5, $I \setminus \mathbb{Z}$ contains rationals X with $\kappa_{3,2}(\sqrt{X}) > 0$. $\square$

Only $E = \emptyset$ is covered. Whether $\mathfrak{B} = \emptyset$ is open (Section 7).

4. POSITIVE SPANNING BY THE BRS FUNCTIONS

The argument behind Fact 2.3 gives more than its statement: a positive spanning property of the values of the BRS functions U_m , $m \equiv 2 \pmod{3}$, on the zero side. It is recorded here because it comes from the same class sums; it gives no information on extra prime atoms (Remark 4.2).

Proposition 4.1 (Positive spanning by the BRS functions). *Let $E \subset (0, \infty) \setminus Z_\zeta$ be finite.*

- (a) *For every real vector $a = (a_e)_{e \in E} \neq 0$ there is $m \equiv 2 \pmod{3}$ with $\sum_{e \in E} a_e U_m(e) > 0$. Equivalently, the vectors $(U_m(e))_{e \in E}$, $m \equiv 2 \pmod{3}$, positively span $\mathbb{R}^E$, and finitely many of them suffice.*
- (b) *For every real vector $a = (a_e)_{e \in E \cup \{0\}} \neq 0$ with $a_0 \geq 0$ there is $m \equiv 2 \pmod{3}$ with $\sum_{e \in E \cup \{0\}} a_e U_m(e) > 0$. Equivalently, the unit vector δ_0 at the point 0 lies in the interior of the convex cone in $\mathbb{R}^{E \cup \{0\}}$ generated by the vectors $(U_m(e))_{e \in E \cup \{0\}}$, $m \equiv 2 \pmod{3}$.*

Proof. Let E' be E in (a) and $E \cup \{0\}$ in (b), and suppose that $\sum_{e \in E'} a_e U_m(e) \leq 0$ for every $m \equiv 2 \pmod{3}$. Put $s_e := \frac{1}{4} + \frac{ie}{2}$ and $\tilde{\varrho}_e := a_e \Xi(e) / (\frac{1}{4} + e^2)$ for $e \in E'$ (so $\tilde{\varrho}_0 = 4a_0 \Xi(0)$), and $c_N := \sum_{e \in E'} \tilde{\varrho}_e \alpha_N(s_e)$. At $s = \frac{1}{2} + ie$ one has $\frac{1}{2}s(s-1) = -\frac{1}{2}(\frac{1}{4} + e^2)$, so $\Gamma_{\mathbb{R}}(s)\zeta(s) = -2\Xi(e)/(\frac{1}{4} + e^2)$, and $\frac{s}{2} = s_e$. Hence (2.2) gives

$$-2\pi m^{1/4} \sum_{e \in E'} a_e U_m(e) = \sum_{d^2 | m} \boldsymbol{\mu}(d) c_{m/d^2} =: c_m^*,$$

and $c_m^* \geq 0$ for every $m \equiv 2 \pmod{3}$. By Möbius inversion, $c_N = \sum_{k^2 | N} c_{N/k^2}^*$. If $N \equiv 2 \pmod{3}$ and $k^2 | N$, then $3 \nmid k$, so $N/k^2 \equiv 2 \pmod{3}$; hence $c_N \geq 0$ for every $N \equiv 2 \pmod{3}$. In (b), $\tilde{\varrho}_0 \geq 0$ because $\Xi(0) > 0$. Fact 2.7, applied with $E_+ = E'$ and with the weight $2a_0$ in place of a_0 (which gives the same $\tilde{\varrho}_0$), yields $\tilde{\varrho}_e = 0$ for every $e \in E'$, and $\Xi(e) \neq 0$ because $e \notin Z_\zeta$; so $a = 0$.

The equivalences are facts about the convex cone $K \subset \mathbb{R}^n$ generated by the vectors in question and its dual cone $K^* = \{a : \langle a, v \rangle \geq 0 \text{ for all } v \in K\}$. In (a) the statement says that $K^* = \{0\}$. Then K is dense, hence equal to $\mathbb{R}^n$, because a convex set and its closure have the same interior; so each of the $2n$ vectors $\pm \delta_e$ is a finite non-negative combination of generators. In (b) the statement, applied to $-a$, says that $a_0 > 0$ for every non-zero $a \in K^*$, and this holds if and only if δ_0 lies in the interior of K . $\square$

Remark 4.2 (Testing admissible pairs). Let (μ, ν) be an admissible pair with μ carried by $Z_\zeta \cup E$, E finite and symmetric, and ν carried by $\mathcal{M}$ (Lemma 2.2 with $\mathcal{X} = \emptyset$), and put $a_e := \mu(\{e\}) \geq 0$. Let $F = \sum_{m \equiv 2 \pmod{3}} x_m U_m$ be a finite combination with $x_m \geq 0$. Then $\mathcal{A}(F) = 0$ by (2.3), since no $m \equiv 2 \pmod{3}$ is a square; F vanishes at every t_ρ ; and $\widehat{F}$ equals x_m at the node $\log m/4\pi$ and 0 at the other nodes. So the explicit formula for the pair gives $\sum_{e \in E} a_e F(e) = -\frac{1}{\pi} \sum_m x_m \nu(\{\frac{\log m}{4\pi}\}) \leq 0$. If $0 \in E$, Proposition 4.1(b), applied to $E_+ \setminus \{0\}$, gives such an F with $F(0) > 0$ and $F = 0$ on $E \setminus \{0\}$; so $a_0 F(0) \leq 0$ and $a_0 = 0$. Then, for each $e_0 \in E_+ \setminus \{0\}$, part (a), applied to $E_+ \setminus \{0\}$, gives such an F with $F(e_0) > 0$ and $F = 0$ on $E \setminus \{0, \pm e_0\}$; so $2a_{e_0} F(e_0) \leq 0$ and $a_{e_0} = 0$. Hence μ is carried by Z_ζ , and $(\mu, \nu) = p_\zeta$ by Lemma 2.2: for admissible pairs, testing with explicit functions recovers Fact 2.3.

These test functions change sign on $\mathbb{R}$: $\int F = \widehat{F}(0) = 0$, because the node $0 = \log 1/4\pi$ belongs to the square $m' = 1$ and $\widehat{U_m}(0) = 0$ for $m \geq 2$. So they are not in the cone of non-negative functions with non-negative transform beyond the gap. In the terms of Lemma 3.3 they are node-positive, but nothing controls the sign of $\widehat{F}$ at a point $\eta \notin \mathcal{M}$, so they give no information on extra prime atoms.

5. COUNTABLY MANY EXTRA ZEROS: AN OUTLINE

Remark 5.1 (Countably many extra zeros). We had proposed to extend Fact 2.3 to countable, locally finite sets E with $\sum_{e \in E} \mu(\{e\}) |\zeta(\frac{1}{2} + ie)| (1 + |e|)^{-1/4+\delta} < \infty$ for some $\delta > 0$, which would need control, uniform in e , of the cusp expansions at the regular points $\frac{2}{3}$ and $\frac{1}{2}$. This is now proved in [4, Theorem 3.8], by a theorem of Astra, under the weaker hypothesis $\sum_{e \in E} \mu(\{e\}) |\zeta(\frac{1}{2} + ie)| (1 + |e|)^{-1/2} < \infty$ (more generally, when the partial sums of $\mu(\{e\}) |\zeta(\frac{1}{2} + ie)|$ over $0 \leq e \leq T$ are $o(\sqrt{T})$) and without local finiteness: the intermediate cusp terms are bounded in aggregate, and a mean-value argument replaces Landau's theorem. Without such a summability condition, for instance for a positive density of extra zeros with masses of order 1, the question remains open.

6. NUMERICAL OBSERVATIONS

The two observations below were computed but not certified; they are not used in any proof.

Numerical observation 6.1 (Rational extras). For each of the 126 rationals $X = a/d \in (\frac{1}{2}, 12) \setminus \mathbb{Z}$ with $2 \leq d \leq 6$, some class (q, b) with $q \in \{3, 5, 7, 11, 13\}$ has computed value $x \kappa_{q,b}(x) > 0$; the smallest such value is 0.004278, at $X = 7/6$, $(q, b) = (13, 6)$. For the 7875 pairs of these values, a non-negative combination of the 17 class vectors with $q \leq 13$ is positive at both points for 7872 pairs, and for the remaining 3 once the classes with $q \in \{17, 19, 23\}$ are added. The values for $q = 3$ are exact Gauss-sum evaluations; for $q \geq 5$ they were obtained by numerical extrapolation of $v^{1/2} T_{q,b}^{(x)}(v)$ as $v \downarrow 3 \cdot 10^{-6}$, with no closed form and no interval certification. Only $X > 4$ corresponds to positions in $[\xi_2, \infty)$. If certified, Theorem 3.4(a) would exclude a lone rational extra atom at each such X when $E = \emptyset$; for admissible pairs that case is now settled by Fact 2.4, so these values concern the class sums only.

Numerical observation 6.2 (Irrational extras at intermediate scales). Let $E = \emptyset$. For each of 12 irrational-type positions (seven shifted prime powers $x^2 = n^2 \pm \delta$ with $n \in \{3, 5, 7, 8, 11\}$ and $\delta \in [0.003, 0.067]$, and $x^2 \in \{4 + (\sqrt{2} - 1), 5 + (\phi - 1), 9 + 1/e, 16 + \sqrt{3}/10, 30 + \sqrt{7}/10\}$, ϕ the golden ratio), the computed function $v^{1/4} T_{3,2}^{(x)}(v)$ reaches a positive value, between +0.10 and +0.81, somewhere in $10^{-9} \leq v \leq 10^{-2.75}$; and for all 15 pairs formed from six of these positions the corresponding linear feasibility problem has positive margin (0.08–0.67). The class sums were computed from the exact cocycle formula with exponentially small terms dropped and ϑ evaluated to 25 digits; they are not interval-certified. If certified, Lemma 2.2 would exclude a lone extra atom (resp. these pairs) at these positions; with $E = \emptyset$ this is now implied by Fact 2.4, so the computation concerns the class sums only.

7. OPEN PROBLEMS

- (1) *Theta class sums.* Is the set $\mathfrak{B}$ of Proposition 3.8 empty? For admissible pairs without extra zeros the question is settled by Fact 2.4, but it remains a question about theta class sums: Theorem 3.4 and Proposition 3.8 leave a closed, nowhere-dense set of positions, and irrational positions are invisible to the leading-order argument (Proposition 3.6). This seems to need theta-sum asymptotics along continued fractions and a Duffin–Schaeffer-type step [2, 5].
- (2) *Extra zeros and extra prime atoms together.* Exclude extra prime mass when μ has finitely many extra zeros and ν is arbitrary. Lemma 3.3 gives one route: a single node-positive test function that vanishes on E and is positive at the position of the atom excludes it.

- (3) *Extra zeros and other classes.* For $E \neq \emptyset$ and $q \geq 5$, the analogue of Fact 2.6 for the non-residue classes $b \bmod q$, which would extend Theorem 3.4(b) to these classes.
- (4) *Infinitely many extra zeros* without the summability condition of Section 5.
- (5) *Certification* of the computations of Numerical observations 6.1 and 6.2.

AI-USE STATEMENT

This note was produced with substantial use of AI. The mathematical arguments, the numerical computations and the text were generated by instances of Claude Opus 5.5 (Anthropic) working as coordinated agents: one set of agents developed the arguments, separate agents acted as independent hostile referees of each component, and the referees' requested repairs were incorporated. The zero-side support theorem and its extension to countably many extra zeros (Theorems 3.6 and 3.8 of the companion paper), which this note cites, were contributed by Astra (OpenAI) and checked by independent referee agents before inclusion. The author initiated and directed this project, designed its verification process, and reviewed the results and the verification record, but did not independently check every step of the written proofs. The author takes full responsibility for the content of this note, including any errors, and welcomes independent verification and corrections.

APPENDIX A. PROOFS

A.1. Structure of pairs with extra atoms.

Proof of Lemma 2.2. Since U_m vanishes on Z_ζ , $\widehat{U_m}(\frac{\log m'}{4\pi}) = \delta_{mm'}$ and (2.3) holds, testing on U_m gives

$$\sum_{e \in E} a_e U_m(e) + \frac{1}{\pi} \left(\nu(\{\frac{\log m}{4\pi}\}) + \sum_{\eta} \beta_{\eta} \widehat{U_m}(\eta) \right) = \frac{1}{\pi} \Lambda(\sqrt{m}) m^{-1/4}.$$

Multiply by $\pi m^{1/4}$. By (2.2) and $\Gamma_{\mathbb{R}}(\frac{1}{2} + it) \zeta(\frac{1}{2} + it) = -2\Xi(t)/(\frac{1}{4} + t^2)$, $\pi m^{1/4} a_e U_m(e) = -\frac{a_e \Xi(e)}{2(\frac{1}{4} + e^2)} \sum_{d^2|m} \mu(d) \alpha_{m/d^2}(s_e)$. The atoms at $\pm e$ give equal terms, because $a_{-e} = a_e$, Ξ is even and $\alpha_N(s_{-e}) = \alpha_N(\overline{s_e}) = \alpha_N(s_e)$; so $\pi m^{1/4} \sum_e a_e U_m(e) = -\sum_{d^2|m} \mu(d) c_{m/d^2}(E)$. With $m^{1/4} \widehat{U_m}(\eta) = \sqrt{x_{\eta}} u_m(x_{\eta})$ we get

$$\ell_m = \Lambda(\sqrt{m}) \mathbf{1}_{\square}(m) + \sum_{d^2|m} \mu(d) \left(c_{m/d^2}(E) - \sum_{\eta} \beta_{\eta} \sqrt{x_{\eta}} B_{m/d^2}(x_{\eta}) \right).$$

Möbius inversion over squares ($f_m = \sum_{d^2|m} \mu(d) g_{m/d^2}$ for all m if and only if $g_N = \sum_{k^2|N} f_{N/k^2}$ for all N) and $\sum_{k^2|N} \Lambda(\sqrt{N/k^2}) \mathbf{1}_{\square}(N/k^2) = \sum_{k|M} \Lambda(M/k) = \log M$ for $N = M^2$ give (2.4). Now let $\mathcal{X} = \emptyset$. If ν lives on $[\xi_2, \infty)$ then $\ell_1 = \ell_2 = \ell_3 = 0$, and (2.4) at $N \leq 3$ gives $c_N = 0$. The remaining statements for $\mathcal{X} = \emptyset$ come from testing on $V_{\rho,j}$, whose transform vanishes on $\mathcal{M}$, with (2.3), and from the uniqueness of ζ 's pair on its own support; they are proved in [4]. Finally, if $(\mu, \nu) \in \mathcal{K}$ then $\ell_m \geq 0$, so $C_N \geq 0$. $\square$

A.2. The theta function near rational points.

Proof of Lemma 2.5. (a) Since $e^{\pi i n^2 r}$ depends only on $n \bmod 2c$, Poisson summation in each residue class gives

$$\vartheta(r+z) = \sum_{\nu \bmod 2c} e^{\pi i \nu^2 r} \sum_{m \in \mathbb{Z}} e^{\pi i (\nu + 2cm)^2 z} = \frac{(z/i)^{-1/2}}{2c} \sum_{\nu \bmod 2c} e^{\pi i \nu^2 r} \sum_{k \in \mathbb{Z}} e^{\pi i k \nu / c} e^{-\pi i k^2 / (4c^2 z)}.$$

The term $k = 0$ is $(z/i)^{-1/2} (2c)^{-1} \sum_{\nu \bmod 2c} e^{\pi i \nu^2 a/c} = g(r) (z/i)^{-1/2}$: if ac is even, $\nu \mapsto e^{\pi i \nu^2 a/c}$ has period c ; if ac is odd, the terms ν and $\nu + c$ cancel. The terms $k \neq 0$ have modulus at most

$e^{-\pi k^2 Y/(4c^2)}$. In a sector $|\operatorname{Re} z| \leq M \operatorname{Im} z$ one has $Y = \operatorname{Im} z/|z|^2 \geq (1 + M^2)^{-1/2}|z|^{-1}$. (b) First, $|\vartheta(r + \delta + i\varepsilon) - \vartheta(r + i\varepsilon)| \leq \sum_n \pi n^2 |\delta| e^{-\pi n^2 \varepsilon} \ll M \varepsilon^2 \varepsilon^{-3/2}$. Next, by Weyl's equidistribution theorem for $n^2 r/2$ [6, Ch. 1, Theorems 2.1 and 3.2], $S(n) := \sum_{|k| \leq n} e^{\pi i k^2 r} = o(n)$. Partial summation gives $\vartheta(r + i\varepsilon) = \sum_{n \geq 0} S(n)(e^{-\pi n^2 \varepsilon} - e^{-\pi(n+1)^2 \varepsilon})$; given $\eta > 0$, split at n_0 with $|S(n)| \leq \eta n$ for $n \geq n_0$ and use $\sum_n n(e^{-\pi n^2 \varepsilon} - e^{-\pi(n+1)^2 \varepsilon}) = \sum_{n \geq 1} e^{-\pi n^2 \varepsilon} \leq (2\varepsilon)^{-1/2}$. (c) Let $\tau = r + i\varepsilon$, $\varepsilon \rightarrow 0^+$. Then $-1/\tau = -1/r + z'$ with $z'/i = \varepsilon r^{-2}(1 + O(\varepsilon))$, so by (a) both sides of $\vartheta(-1/\tau) = (\tau/i)^{1/2} \vartheta(\tau)$ are $\varepsilon^{-1/2}$ times a constant plus $o(\varepsilon^{-1/2})$: the constant is $g(-1/r)|r|$ on the left and $(r/i)^{1/2} g(r)$ on the right. As $(r/i)^{1/2} = |r|^{1/2} e^{-i\pi \operatorname{sgn}(r)/4}$, the claim follows. $\square$

A.3. Class sums of an extra atom.

Proof of Proposition 3.2. Fix $r = 2j/q$ with $0 \leq j < q$. Reduce r to ∞ by the following adaptation to cusps of the reduction algorithm of [1, §3.1] (which is stated for points of $\mathbb{H}$): translate by an even integer into $[-1, 1]$, then apply S , and repeat. On reduced fractions a/c the translations preserve the parities of a and c and the denominator, and $S : a/c \mapsto -c/a$ exchanges the types (even, odd) and (odd, even) and lowers the denominator, since $|a| < c$ in $[-1, 1]$ (the value ± 1 , of type (odd, odd), does not occur). So the procedure ends at $0/1$ followed by a final S ; it gives a word $\gamma_r = g_m \cdots g_1$ with $g_m = S$, $\gamma_r r = \infty$, and $w_{m-1} r = 0$. As $r = 2j/q$ and $0/1$ have type (even, odd), the number of S 's in w_{m-1} is even, so $\chi(\gamma_r) = -1$ and $\chi(w_{m-1}) = +1$. Every other k with $g_k = S$ has $\rho_k := w_{k-1} r \in \mathbb{Q} \setminus \{0\}$. Let $\tau = r + iv$. Each w_{k-1} is a Möbius map, finite at r , so $w_{k-1} \tau = \rho_k + i\lambda_k v + O(v^2)$ with $\lambda_k := w'_{k-1}(r) > 0$; and $\operatorname{Im} \gamma_r \tau = 1/(c_r^2 v)$ where $c_r \neq 0$ is the lower-left entry of γ_r . Apply (2.5) to G_x .

The term $G_x|_{\gamma_r}$. It equals $-\vartheta(\tau) G_x(\gamma_r \tau) / \vartheta(\gamma_r \tau) = -B_0(x) \vartheta(\tau) + O(v^{-1/2} e^{-\pi/(c_r^2 v)})$.

The term $k = m$. Here $w := w_{m-1} \tau \rightarrow 0$ non-tangentially. Inverting both theta functions in $\mathfrak{p}_x(w)$, $\vartheta(x^2 w) = x^{-1} (w/i)^{-1/2} \vartheta(-1/(x^2 w))$ and $x^{-1} \vartheta(w/x^2) = (w/i)^{-1/2} \vartheta(-x^2/w)$, gives $\mathfrak{p}_x(w) = \frac{1}{2x} (w/i)^{-1/2} - \frac{1}{2} + O(|w|^{-1/2} e^{-c/|w|})$. Since $(w/i)^{-1/2} / \vartheta(w) = 1/\vartheta(-1/w) = 1/\vartheta(\gamma_r \tau)$,

$$(\mathfrak{p}_x|_{w_{m-1}})(\tau) = \frac{\vartheta(\tau)}{\vartheta(w)} \mathfrak{p}_x(w) = \frac{1}{2x} \vartheta(\tau) - \frac{1}{2} \frac{\vartheta(\tau)}{\vartheta(w_{m-1} \tau)} + O(v^{-1/2} e^{-c/v}),$$

and the middle term is bounded: $\vartheta(\tau)/\vartheta(w_{m-1} \tau)$ is a product of factors $(w_{k-1} \tau/i)^{-1/2}$ over the S 's in w_{m-1} , evaluated at points tending to $\rho_k \neq 0$.

The terms $k < m$ with $g_k = S$. Here $(\mathfrak{p}_x|_{w_{k-1}})(\tau) = \chi(w_{k-1}) j_k(\tau) \mathfrak{p}_x(w_{k-1} \tau)$, where $j_k(\tau) := \vartheta(\tau)/\vartheta(w_{k-1} \tau)$ is analytic near r with a value $j_k \neq 0$ there. In $\mathfrak{p}_x(w_{k-1} \tau)$ the terms -1 and $(w_{k-1} \tau/i)^{-1/2}$ are bounded, while $\vartheta(x^2 w_{k-1} \tau)$ and $\vartheta(w_{k-1} \tau/x^2)$ are theta values near the real points $x^2 \rho_k$ and ρ_k/x^2 , approached in the manner of Lemma 2.5(a),(b). If $x^2 \notin \mathbb{Q}$ these points are irrational and the terms are $o(v^{-1/2})$. If $x^2 \in \mathbb{Q}$, Lemma 2.5(a) gives $\vartheta(x^2 w_{k-1} \tau) = g(x^2 \rho_k)(x^2 \lambda_k v)^{-1/2} + O(v^{1/2})$ and $\vartheta(w_{k-1} \tau/x^2) = g(\rho_k/x^2)(\lambda_k v/x^2)^{-1/2} + O(v^{1/2})$.

Summation over the cusps. The coefficient of $\vartheta(\tau)$ is $-(B_0(x) - \frac{1}{2x})$ at every cusp $2j/q$, and $\sum_{j \bmod q} e^{-2\pi i j b/q} \vartheta(\frac{2j}{q} + iv) = q \sum_{n^2 \equiv b(q)} e^{-\pi n^2 v} = 0$ since b is a non-residue. All other terms are $O(1)$, exponentially small, or theta values at the sampled points. Hence $T_{q,b}^{(x)}(v) = \kappa_{q,b}(x) v^{-1/2} + o(v^{-1/2})$ with $\kappa_{q,b}(x) = 0$ if $x^2 \notin \mathbb{Q}$, and with an error $O(1)$ and

$$(A.1) \quad \kappa_{q,b}(x) = \frac{1}{q} \sum_{j \bmod q} e^{-2\pi i j b/q} \sum_{k < m(j), g_k = S} \chi(w_{k-1}) j_k \frac{\lambda_k^{-1/2}}{2} \left[\frac{g(x^2 \rho_k)}{x} + g(\rho_k/x^2) \right]$$

if $x^2 \in \mathbb{Q}$ (all quantities depending on j).

The case $(q, b) = (3, 2)$. For $j = 0$ the word is S and there is no term with $k < m$. For $j = 1$ the word is ST^2ST^2S (Section 2.4) and the terms $k < m$ are $k = 1$ ($w_0 = 1$, $\rho = \frac{2}{3}$, $\chi = 1$,

$j_1 = 1, \lambda = 1$) and $k = 3$ ($w_2 = T^2 S, \rho = \frac{1}{2}, \chi = -1, j_3 = (\frac{2}{3i})^{-1/2} = (\frac{3}{2})^{1/2} e^{i\pi/4}, \lambda = (\frac{3}{2})^2$). With $X = x^2$ their contribution is

$$K := \frac{1}{2} \left[\frac{g(\frac{2X}{3})}{x} + g(\frac{2}{3X}) \right] - \frac{e^{i\pi/4}}{\sqrt{6}} \left[\frac{g(\frac{X}{2})}{x} + g(\frac{1}{2X}) \right].$$

By Lemma 2.5(c), $g(\frac{2}{3X}) = (\frac{3X}{2})^{-1/2} e^{i\pi/4} g(-\frac{3X}{2})$ and $g(\frac{1}{2X}) = (2X)^{-1/2} e^{i\pi/4} g(-2X)$, which turns K into the expression in Proposition 3.2. For $j = 2, \frac{4}{3} \equiv -\frac{2}{3} \pmod{2}$ and the $B_N(x)$ are real, so $G_x(\frac{4}{3} + iv) = \overline{G_x(\frac{2}{3} + iv)}$ and the contribution is $\overline{K}$. With $e^{-4\pi i/3} = \omega$ and $e^{-8\pi i/3} = \bar{\omega}$ we get $\kappa_{3,2}(x) = \frac{1}{3}(\omega K + \bar{\omega} \overline{K}) = \frac{2}{3} \operatorname{Re}(\omega K)$.

Integers. If $X = m \in \mathbb{N}$, then $B_N(\sqrt{m}) = \sum_{k \geq 1} b_N(k\sqrt{m}) = \#\{k \geq 1 : k^2 m = N\}$, so $G_x = \frac{1}{2}(\vartheta(m\tau) - 1)$ and $T_{3,2}^{(x)}(v) = \sum_{k \geq 1, k^2 m \equiv 2 \pmod{3}} e^{-\pi k^2 m v}$, which is $\frac{1}{3}x^{-1}v^{-1/2} + O(1)$ if $m \equiv 2 \pmod{3}$ and 0 otherwise. Evaluating K directly (for instance $m = 5$: $g(\frac{10}{3}) = -i/\sqrt{3}$, $g(-\frac{15}{2}) = g(\frac{5}{2}) = \frac{1+i}{2}$, $g(-10) = 1$, so $K = -i/(\sqrt{3}x)$) gives the same values. $\square$

A.4. A closed form on a dense family.

Proof of Lemma 3.5. Let $G(c; n) := \sum_{k \bmod n} e^{2\pi i c k^2/n}$. We use the standard facts on quadratic Gauss sums [3, §3.4]: $G(c; n) = (c/n)G(1; n)$ for odd n and $(c, n) = 1$, with $G(1; n) = \sqrt{n}$ or $i\sqrt{n}$ according as $n \equiv 1$ or $3 \pmod{4}$; $G(c; n_1 n_2) = G(c n_2; n_1)G(c n_1; n_2)$ for coprime n_1, n_2 ; and $G(c; 4) = 2(1 + i^c)$ for odd c . Since a is odd, $3 \nmid a, d \nmid a$ and $d \neq 3$, the four points $\frac{2a}{3d}, -\frac{2a}{d}, \frac{a}{2d}, -\frac{3a}{2d}$ are in lowest terms with even numerator or even denominator. Summing over $n \bmod 2c$ when the denominator c is even, one finds

$$g(\frac{2X}{3}) = \frac{G(a; 3d)}{3d}, \quad g(-2X) = \frac{G(-a; d)}{d}, \quad g(\frac{X}{2}) = \frac{G(a; 4d)}{4d}, \quad g(-\frac{3X}{2}) = \frac{G(-3a; 4d)}{4d}.$$

Now $d \equiv 1 \pmod{12}$ gives $(-1/d) = (3/d) = (d/3) = 1$, and $a \equiv 2 \pmod{3}$ gives $(a/3) = -1$. Hence $G(a; 3d) = G(ad; 3)G(3a; d) = -i\sqrt{3}(a/d)\sqrt{d}$, $G(-a; d) = (a/d)\sqrt{d}$, and, as $i^{ad} = i^{-3ad} = i^a$, $G(a; 4d) = G(4a; d)G(ad; 4) = 2(1 + i^a)(a/d)\sqrt{d} = G(-12a; d)G(-3ad; 4) = G(-3a; 4d)$. So $g(-\frac{3X}{2}) = g(\frac{X}{2})$, $g(\frac{2X}{3}) = -i(a/d)/\sqrt{3d}$, $g(-2X) = (a/d)/\sqrt{d}$, and $K = -i(a/d)/(x\sqrt{3d})$. Then $\operatorname{Re}(\omega K) = \frac{\sqrt{3}}{2} \cdot \frac{(a/d)}{x\sqrt{3d}}$, which gives the formula. For density, fix an interval $(\alpha, \alpha') \subset (0, \infty)$. For a prime $d \equiv 1 \pmod{24}$ there are $(\alpha' - \alpha)d/6 + O(1)$ integers $a \equiv 5 \pmod{6}$ with $a/d \in (\alpha, \alpha')$, of which $O(1)$ are divisible by d ; writing $a = 6a' + 5$, the sum of (a/d) over them is a sum of a non-principal character over an interval of consecutive integers $a' + 5 \cdot \bar{6}$, hence $O(\sqrt{d} \log d)$ by the Pólya–Vinogradov inequality [3, Ch. 12]. By Dirichlet's theorem there are arbitrarily large such primes d , and for large d some of these a have $(a/d) = +1$. $\square$

A.5. Continuity in x .

Proof of Lemma 3.7. Let $\phi_x(z) = \frac{1}{2}(\vartheta(x^2 z) - 1)$ and fix $0 < a < b$. For $x \in [a, b]$ and $z \in \mathbb{H}$, $|\phi_x(z)| \leq \sum_{n \geq 1} e^{-\pi n^2 a^2 \operatorname{Im} z} \leq (2a)^{-1}(\operatorname{Im} z)^{-1/2}$, and $x \mapsto \phi_x(z)$ is continuous.

By [1, Proposition 3.3], the function $G_x = F_{1/2}^-(\cdot, \phi_x)$ is the analytic continuation to $\mathbb{H}$ of the integral $\frac{1}{2} \int_{\ell_0} \mathcal{K}(\tau, z) \phi_x(z) dz$ ($\tau \in \mathfrak{F}$), where $\mathfrak{F} = \{|\operatorname{Re} \tau| < 1, |\tau| > 1\}$, ℓ_0 is the upper unit semicircle from -1 to 1 , and $\mathcal{K} = \mathcal{K}_{1/2}^-$ is the modular kernel of [1, §3.3]. For fixed τ , $z \mapsto \mathcal{K}(\tau, z)$ is meromorphic in $\mathbb{H}$ with poles only on $\Gamma_\vartheta \tau$, and it decays like $e^{-c/\operatorname{Im} z}$ as $z \rightarrow \pm 1$ non-tangentially. The continuation is described in the proof of that proposition: the same integral defines it across the vertical sides $\operatorname{Re} \tau = \pm 1$ of $\mathfrak{F}$ (no point of $\Gamma_\vartheta \tau$ lies on ℓ_0 for such τ); and near ℓ_0 , $G_x = \frac{1}{2} \int_{\ell_1} \mathcal{K}(\cdot, z) \phi_x(z) dz + \phi_x$ on the region between a path ℓ_1 from -1 to 1 above ℓ_0 , meeting the real axis non-tangentially, and its image under S . Hence every point w

of $\overline{\mathfrak{F}} \cap \mathbb{H}$ has a neighbourhood on which $G_x(w)$ is given by one of these integrals (plus $\phi_x(w)$), with an integrand dominated, for all $x \in [a, b]$, by $|\mathcal{K}(w, z)|(2a)^{-1}(\operatorname{Im} z)^{-1/2}$, which is integrable along the path. By dominated convergence, $x \mapsto G_x(w)$ is continuous on $[a, b]$ for every such w .

For general $\tau \in \mathbb{H}$ choose $\gamma \in \Gamma_\vartheta$ with $\gamma\tau \in \overline{\mathfrak{F}}$ [1, §3.1]. By (2.5), $G_x(\tau) = \chi(\gamma)^{\frac{\vartheta(\tau)}{\vartheta(\gamma\tau)}} G_x(\gamma\tau) + \sum_{k: g_k=S}(\mathfrak{p}_x|w_{k-1})(\tau)$. The first term is continuous in x by the previous paragraph, and the finitely many others are values of $\mathfrak{p}_x$, which is continuous in x , at points independent of x , times factors independent of x . $\square$

REFERENCES

- [1] Andriy Bondarenko, Danylo Radchenko, and Kristian Seip, *Fourier interpolation with zeros of zeta and L -functions*, Constr. Approx. **57** (2023), no. 2, 405–461, arXiv:2005.02996, doi:10.1007/s00365-022-09599-w.
- [2] R. J. Duffin and A. C. Schaeffer, *Khintchine’s problem in metric Diophantine approximation*, Duke Math. J. **8** (1941), no. 2, 243–255, doi:10.1215/S0012-7094-41-00818-9.
- [3] Henryk Iwaniec and Emmanuel Kowalski, *Analytic number theory*, American Mathematical Society Colloquium Publications, vol. 53, American Mathematical Society, Providence, RI, 2004, doi:10.1090/coll/053.
- [4] Nic Johns, *Positive solutions of the explicit formula for $\zeta(s)$: near-criticality and uniqueness*, Companion paper, in preparation, 2026.
- [5] Dimitris Koukoulopoulos and James Maynard, *On the Duffin–Schaeffer conjecture*, Ann. of Math. (2) **192** (2020), no. 1, 251–307, arXiv:1907.04593, doi:10.4007/annals.2020.192.1.5.
- [6] L. Kuipers and H. Niederreiter, *Uniform distribution of sequences*, Pure and Applied Mathematics, Wiley-Interscience, New York–London–Sydney, 1974.
- [7] Danylo Radchenko and Maryna Viazovska, *Fourier interpolation on the real line*, Publ. Math. Inst. Hautes Études Sci. **129** (2019), 51–81, arXiv:1701.00265, doi:10.1007/s10240-018-0101-z.
- [8] André Weil, *Sur les “formules explicites” de la théorie des nombres premiers*, Comm. Sém. Math. Univ. Lund [Medd. Lunds Univ. Mat. Sem.] **Tome supplémentaire** (1952), 252–265.